CORE JAVA — PRACTICAL EXAMINATION

Coder Army | Java Full Course 2026

■ Time: 90 Minutes | ■ Solve on IDE | ■ Total Marks: 50 | ■ One .java file per coding question

INSTRUCTIONS: (1) Attempt ALL sections. (2) MCQ — circle one option only, 0.5 marks each, no negative marking. (3) Predict Output — write the exact output, no partial marks for wrong output. (4) Coding — each question in a separate .java file, must compile and run without errors. (5) Bitwise and Shift operators are NOT included. (6) Push all coding files to GitHub folder practical-exam-lec1-7 after the exam.

SECTION A — Multiple Choice Questions

10 Questions × 0.5 Marks = 5 Marks | Circle the correct option.

Q1. Which of the following correctly declares a long variable in Java? **[0.5]**

(a) long x = 10;
(b) long x = 10L;
(c) Both (a) and (b) are valid
(d) long x = 10.0;

Q2. What is the output of: int a = 10; int b = 3; System.out.println(a % b); **[0.5]**

(a) 3
(b) 0
(c) 1
(d) 3.33

Q3. int j = 5; int k = j++; — What are the values of j and k after this line? **[0.5]**

(a) j = 5, k = 5
(b) j = 6, k = 5
(c) j = 6, k = 6
(d) j = 5, k = 6

Q4. Which of the following will cause a compile-time error? **[0.5]**

(a) int a = 'A';
(b) double d = 5;
(c) float f = 3.14;
(d) long l = 100L;

Q5. What is the result of: boolean x = (5 > 3) && (10 < 5); **[0.5]**

(a) true
(b) false
(c) 1
(d) Compile error

Q6. In an if-else-if ladder, what happens once the FIRST true condition is found? **[0.5]**

(a) All remaining conditions are still evaluated
(b) Only the very next condition is also evaluated
(c) All remaining conditions and blocks are skipped entirely
(d) A runtime exception is thrown

Q7. What happens if no case matches in a switch and there is NO default? **[0.5]**

(a) Compile-time error
(b) Runtime exception
(c) The first case executes automatically
(d) Nothing — execution continues after the switch block

Q8. int x = 3; switch(x) { case 3: System.out.println("Three"); case 4: System.out.println("Four"); } — What prints? **[0.5]**

(a) Three
(b) Four
(c) Three followed by Four
(d) Nothing

Q9. Which data type CANNOT be used as a switch expression in Java? **[0.5]**

(a) int
(b) char
(c) boolean
(d) String

Q10. What does the following print? int a = 5; System.out.println(a++); System.out.println(a); **[0.5]**

(a) 5 then 5
(b) 6 then 6
(c) 5 then 6
(d) 6 then 5

SECTION B — Predict the Output

10 Questions × 1 Mark = 10 Marks | Write the exact output. No partial marks.

Q11. What is the exact output? **[1]**

```
int a = 17, b = 5;
System.out.println(a / b);
System.out.println(a % b);
```

Output: ______________________________

Q12. What is the exact output? **[1]**

```
int a = 5;
a += 3;
a *= 2;
System.out.println(a);
```

Output: ______________________________

Q13. What is the exact output? **[1]**

```
int j = 7;
j++;
++j;
int k = j++;
System.out.println(j + " , " + k);
```

Output: ______________________________

Q14. What is the exact output? **[1]**

```
int a = 5, b = 10;
System.out.println(a == b);
System.out.println(a != b);
System.out.println(a <= b);
```

Output: ______________________________

Q15. What is the exact output? **[1]**

```
boolean x = (10 > 5) && (3 < 1);
boolean y = (10 > 5) || (3 < 1);
System.out.println(x);
System.out.println(y);
```

Output: ______________________________

Q16. What is the exact output? **[1]**

```
int marks = 72;
if (marks >= 90) System.out.println("A");
else if (marks >= 75) System.out.println("B");
else if (marks >= 60) System.out.println("C");
else if (marks >= 50) System.out.println("D");
else System.out.println("F");
```

Output: ______________________________

Q17. What is the exact output? [1]

```
int a = 1;
if (a > 0)
    if (a > 5)
        System.out.println("Greater than 5");
    else
        System.out.println("Between 1 and 5");
else
    System.out.println("Not positive");
```

Output: ____

Q18. What is the exact output? [1]

```
int x = 2;
switch (x) {
    case 1: System.out.println("One");
    case 2: System.out.println("Two");
    case 3: System.out.println("Three");
    default: System.out.println("Default");
}
```

Output: ____

Q19. What is the exact output? [1]

```
int x = 2;
switch (x) {
    case 1: System.out.println("One"); break;
    case 2: System.out.println("Two"); break;
    case 3: System.out.println("Three"); break;
    default: System.out.println("Default");
}
```

Output: ____

Q20. What is the exact output? [1]

```
int a = 4, b = 3, c = 2;
int result = a * b + c * b - a / b;
System.out.println(result);
```

Output: ____

SECTION C — Coding Questions

5 Questions × 3 Marks = 15 Marks | Write complete Java programs. One .java file each.

Q21. Declare int a = 300 and cast it to byte. Declare double d = 9.75 and cast it to int. Declare int x = 65 and cast it to char. Print all original and casted values with labels. Add a comment for each cast explaining what happened. **[3 marks]**

Q22. Declare int x = 10. Apply these compound operators in order and print x after each step with a label: x += 5 x *= 2 x -= 8 x /= 3 x %= 4 **[3 marks]**

Q23. Declare int a = 25, b = 10, c = 15. Using if-else (not a ladder), check if num is divisible by BOTH 3 and 5 using num = 15. Print 'FizzBuzz' or 'Not FizzBuzz'. Then find the largest among a, b, c using nested if-else and print it. **[3 marks]**

Q24. Declare int marks = 82. Using an if-else-if ladder, print the grade: A (>= 90) | B (>= 75) | C (>= 60) | D (>= 50) | F (below 50) Then declare int income = 750000 and print its tax slab using another if-else-if: 0–250000 → No Tax | 250001–500000 → 5% | 500001–1000000 → 20% | above 1000000 → 30% **[3 marks]**

Q25. Declare int op = 3, a = 12, b = 4. Using switch-case perform: 1 = addition | 2 = subtraction | 3 = multiplication | 4 = division | default = Invalid Store the result in a variable, print it after the switch ends. Then declare int month = 4 and use a second switch with fall-through to print its number of days. **[3 marks]**

SECTION D — Logic Building

2 Questions × 5 Marks = 10 Marks | These combine multiple topics.

Q26. Write a complete Java program for the following: (a) Declare int a = 5, b = 10. Compute and print all five arithmetic results (sum, difference, product, quotient, remainder) with labels. (b) Using if-else, print which of a or b is larger. (c) Using nested if-else, classify b as: positive-even, positive-odd, negative-even, or negative-odd. (d) Add a comment at the top of the file naming all three topics this question covers. **[5 marks]**

Q27. Write a complete Java program for the following: (a) Declare int year = 2100. Check the FULL leap year rule: Divisible by 4 AND (NOT divisible by 100 OR divisible by 400). Print 'Leap Year' or 'Not a Leap Year'. (b) Declare int day = 3. Using switch-case print the day name and whether it is a Weekday or Weekend. (c) In comments, explain step by step why 2100 fails the full leap year rule. **[5 marks]**

Section	Topic	Questions	Marks
A	Multiple Choice	10	5
B	Predict the Output	10	10
C	Coding	5	15
D	Logic Building	2	10
	TOTAL	**27**	**50**

— End of Paper —

Push all .java files to GitHub in folder: practical-exam-lec1-7